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A Tale of Tails: Model Collapse as a Change of Scaling Laws

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Abstract

As AI model size grows, neural *scaling laws* have become an important tool to predict the improvements of large models when increasing capacity and the size of original (human or natural) training data. Yet, the widespread use of popular models means that the ecosystem of online data and text will co-evolve to progressively contain increased amounts of synthesized data. In this paper we ask: *How will the scaling laws change in the inevitable regime where synthetic data makes its way into the training corpus?* Will future models, still improve, or be doomed to degenerate up to total *(model) collapse*? We develop a theoretical framework of model collapse through the lens of scaling laws. We discover a wide range of decay phenomena, analyzing loss of scaling, shifted scaling with number of generations, the "un-learning" of skills, and grokking when mixing human and synthesized data. Our theory is validated by large-scale experiments with a transformer on an arithmetic task and text generation using the large language model `Llama2`.

1. Introduction

Groundbreaking advances in generative AI algorithms for text, images and code are ushering in the "synthetic data age": increasingly we consume data generated by large scale models like GPT4 (Achiam et al., 2023), Stable Diffusion (Rombach et al., 2022) and their successors. A growing number of synthetic data generated with these models starts to populate the web, often indistinguishable from "real" data. Evidence suggests that AI-generated content has contaminated the LAION-5B dataset (Alemohammad et al., 2023) and has been utilized by crowdworkers (Veselovsky et al., 2023). Remarkably, as of the current assessment, ChatGPT contributes to 0.1% of the total words generated daily by the global population (Altman, 2024).

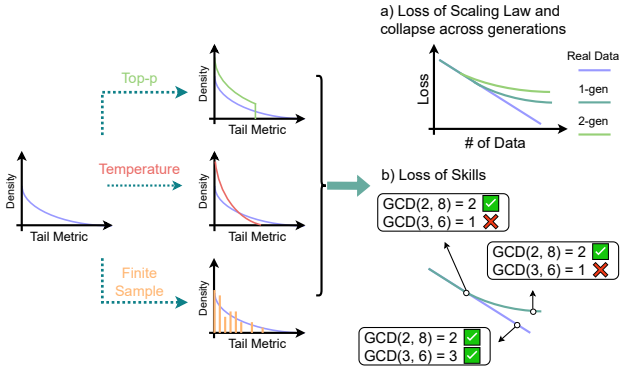

Figure 1. Top-p^{inf} (nucleus) sampling, temperature scaling of LLM generation, and finite sample bias lead to truncated or narrowed "tails" (left side), causing loss of scaling laws (top right) and loss of skills (bottom right). Here we visualize calculating the greatest common divisor (GCD) with answer 2 and 3 as two skills.

At the same time a key driver behind the current success of large models is their consumption of massive amount of web-scale data for training. The improvements of larger models are governed by scaling laws in which error falls off as a power in the size of training data; and the emergence of new skills seems tightly linked to covering increased scales of training data. Our understanding of what the future holds in a world were models are trained on other models (or their own) synthesized data is only at its beginning, but some works indicate the possibility of complete collapse of learning, so called model collapse[1].

Scaling Laws. In many domains of machine learning including speech, translation, vision, video, and mathematical problem solving, empirically observed neural scaling laws (Hestness et al., 2017; Rosenfeld et al., 2020; Kaplan et al., 2020; Hoffmann et al., 2022; Gordon et al., 2021; Henighan et al., 2021; Aghajanyan et al., 2023) demonstrate that test error often falls off as a power law with the amount of training data, model size, and compute. Theoretically, scaling laws have been derived in a variety of settings (e.g. Hutter (2021); Cabannes et al. (2023) for "LLM-like" models).

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[1]Not to be confused with neural collapse which refers to clustering of last-layer features at the end of training (Papyan et al., 2020)

Scaling laws are intimately related to the emergence of abilities (Wei et al., 2022) in larger models, that are not present in smaller ones; and to skills that appear with decreasing loss (Gordon et al., 2021; Arora & Goyal, 2023). This bolsters the now common narrative that "scaling is all you need".

Model Collapse. Current LLMs (Devlin et al., 2018; Liu et al., 2019; Brown et al., 2020; Touvron et al., 2023), including GPT-4 (Achiam et al., 2023), were trained on predominantly human-generated text; similarly, diffusion models like DALL-E (Ramesh et al., 2021), Stable Diffusion (Rombach et al., 2022), Midjourney (Midjourney, 2023) are trained on web-scale image datasets. These training corpora already potentially exhaust all the available clean data on the internet. A growing number of synthetic data generated with these increasingly popular models starts to populate the web, often indistinguishable from "real" data. This proportion is expected to grow, leading us to anticipate a future where AI-generated data predominates. We have thus already entered a future where our training corpora are irreversibly mixed with synthetic data, and this situation is likely to worsen. Recent works call attention to the potential dramatic deterioration in the resulting models, an effect referred to as *"model collapse"* (Shumailov et al., 2023). Facets of this phenomenon have been demonstrated *empirically* in various settings (LeBrun et al., 2021; Hataya et al., 2023; Martínez et al., 2023a;b; Bohacek & Farid, 2023; Briesch et al., 2023; Guo et al., 2023). Theoretically, a few works are emerging to analyze the effect of iterative training on self-generated (or mixed) data: (Shumailov et al., 2023) coin the term *"model collapse"* to characterize complete reversion to the mean, Alemohammad et al. (2023) analyze *"self-consuming loops"*, Bertrand et al. (2023) show that iterative synthetic training leads to a *"clueless generator"*, and Dohmatob et al. (2024) analyze n-fold generation with kernel regression.

With these first warning signs in place, we thus ask:

> *How is the current scaling paradigm affected by synthetic data in the training corpus?*

To this end, we carefully zoom into the scaling behavior of LLM-style models. Theoretical derivations of scaling laws always assume a heavy-tailed distribution (power-law, aka Zipf) on the input features ("heavy tail in, power scaling law out"). This distribution is of the form

$$p_i \propto i^{-\beta}, \quad i = 1, 2, \ldots \tag{1}$$

Such distributions are ubiquitous in natural datasets, from Zipf's law (Zipf, 1935) in distribution of word frequencies, to biological data, earthquake magnitudes, financial data etc. - this is the data being consumed by large models at scale, like LLMs. But what distribution do AI-models generate when trained on such data? Figure 2 provides an empirical answer for a large scale LLM (Llama2-7B) and a transformer model trained on an arithmetic task. Regenerating heavy-tailed data affects the distribution in two possible ways: (1) "Cutting off" the tail of the distribution and/or (2) "Narrowing" the tail (see Figure 1 for a cartoon illustration). The mechanisms leading to this, apart from finite sampling bias (as already proposed in Shumailov et al. (2023) - see Section 2 for a derivation in the Zipf-setting), stem from deliberate choices in the generation algorithms of the models: in LLMs via truncated next token prediction at inference (e.g. selecting more likely tokens via *top-*p^{inf} or *top-*k^{inf} truncation, concentrating the probability distribution by lowering the temperature τ); in vision models like GANs via truncation or in diffusion models through guidance.

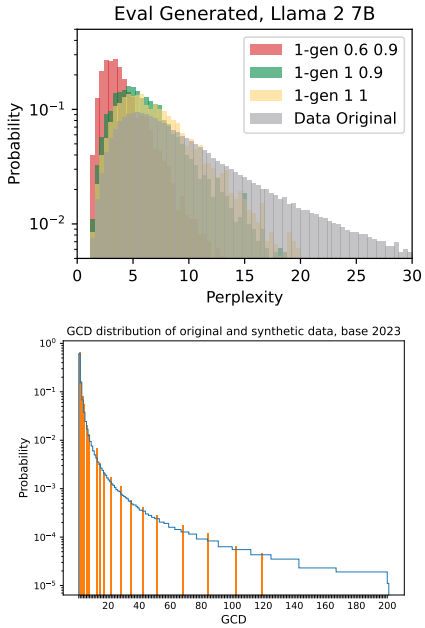

Figure 2. **Tails of AI-generated data: Top.** Perplexity diagram of the Wikitext-103 test set, measured with `Llama2-7B` as the anchor model. We query the Wikitext-finetuned `Llama2-7B` to generate AI data, which is compared to the original set. Perplexity is calculated solely for the generated positions in both the AI and original datasets. AI data is generated for various settings of (p^{inf}, τ). **Bottom.** Distribution of greatest common divisors (GCD) of pairs of random integers (original data (blue) scaling as $p(GCD = k) \propto k^{-2}$). A transformer is trained on this task on $300M$ samples and used as a generator on a test set of randomly sampled integer pairs, giving the truncated GCD distribution.

Summary of Main Contributions. We present a high-level summary of our main theoretical contributions, some of which are highlighted in Figure 3. We empirically verify these theoretical predictions (see Figure 4): (1) in large-scale experiments on an LLM, fine-tuning Llama2-7B (Touvron et al., 2023) on an approximately $2M$ sample dataset from Wikitext-103 and (2) for transformer models trained to predict the greatest common divisor (Charton, 2023).

Assuming a true distribution as in Equation (1), consider

training a model on T AI data-generated data. The synthesized data amounts to a version of the true data distribution with the tail cut at some finite rank k or the tail narrowed to a smaller exponent. Our main findings are as follows.

(1) A Double Scaling Law. We establish new scaling laws that explain model collapse in simplified (non-autoregressive) LM (Hutter, 2021) and toy bigram LLMs (refer to Theorems 2.1 and 4.2)[2]

$$E_{test} \asymp T^{-c} + k^{-c'}. \tag{2}$$

or equivalently (refer to Corollary 2.2), for finite-sample induced cut-off $k = k(T_0)$ when the generating model is trained on T_0 amount of data, $E_{test} \asymp T^{-c} + T_0^{-c''}$, where the exponents c, c', c'' only depend on the tail behavior of the true distribution. This result is illustrated in Figure 3.

For AI-"tail-narrowing", when data remains heavy-tailed, with a smaller exponent $\beta' \in (1, \beta)$, the downstream Hutter LLM will scale as (Corollary 2.3)

$$E_{test} \asymp T^{-(\beta-1)/\beta'}. \tag{3}$$

(2) A Triplet Scaling Law for Memory-Limited Models. We consider a simple associative memory model studied in (Cabannes et al., 2023), and establish (Theorem 5.1) a new scaling law of the form

$$E_{test} \asymp T^{-c} + d^{-c_q} + k^{-c'}, \tag{4}$$

where d is the embedding dimension, and serves a s proxy for model capacity; the exponent c_q depends both on β and the particular algorithm q used to update the memories in the model during training.

(3) Model Collapse over Multiple Generations. For n-fold recursion of AI data-generation (11), where each generation of the model consumes data produced by the previous generation, we establish a universality principle of the form

$$E_{test} = E_{test}^{clean} + n \times \text{ new scaling terms}, \tag{5}$$

where E_{test}^{clean} is the usual test error of the model trained on clean data (not AI-generated). This means that in Equations (2) and (4) for example, the $k^{-c'}$ is replaced by $nk^{-c'}$. One possible interpretation of this multiplicative degradation is that, over time (i.e as the number of generations becomes large), the effect of large language models (like ChatGPT) in the wild will be a pollution of the web to the extend that learning will be impossible. We note that the multiplicative degradation in scaling with the number of generations n is analogous to what has been shown in (Dohmatob et al., 2024) in the context of kernel ridge regression.

[2]The notation $f(T) \lesssim g(T)$ means that $f(T) \leq Cg(T)$ for sufficiently large T and an absolute constant C, while $f(T) \asymp g(T)$ means $f(T) \lesssim g(T) \lesssim f(T)$.

(4) Mitigation Strategies. In Theorem 3.2 we show that mixing AI-generated data with even a small amount of clean data mitigates model collapse by introducing a grokking phenomenon. The length of the plateau is of order k^{β}/π, where π is the proportion of training data which is from the true distribution (i.e clean data). When $\pi = 0$ (i.e only AI-generated data available), this plateau goes on forever (as in (2) and (4)). When $\pi > 0$, however small, the plateau finally halts, and the error continues to decrease à la T^{-c}. This grokking phenomenon holds in the setting of *deterministic* ground truth labels (like in the models of Hutter (2021); Cabannes et al. (2023)). For transformer models, such deterministic settings are found for instance in arithmetic tasks, and we demonstrate it empirically in our GCD transformer experiments. The grokking effect becomes attenuated in probabilistic settings, where it can lead to an S-shaped learning curve (see Figure 19). We also identify regimes where adding AI data can be beneficial and discuss ways to curate "tail" data to mitigate AI-data effects.

Related Work. Theoretically, scaling laws have been derived in various settings: for non-parametric models (Schmidt-Hieber, 2017; Suzuki, 2019; Bordelon et al., 2020), in the kernel regime under the Gaussian design (Spigler et al., 2020; Cui et al., 2021; 2022; 2023; Maloney et al., 2022), or in memorization-like settings with discrete data (Hutter, 2021; Debowski, 2023; Michaud et al., 2023). Taking finite model capacity and optimization into account, Cabannes et al. (2023) recently proved scaling laws in constraint-capacity associative memories, and our Triplet Scaling Law builds on this work.

Less than a handful of works begin to provide theoretical explanations for the behavior of models in the "synthetic data age". (Shumailov et al., 2023) attribute model collapse to two mechanisms: a finite sampling bias cutting off low-probability "tails", thus leading to more and more peaked distributions and function approximation errors; they theoretically analyze the (single) Gaussian case and provide empirical evidence for VAEs, Gaussian mixtures and the OPT language model (125M parameters). In the context of vision models, Alemohammad et al. (2023) analyze *"self-consuming loops"* by introducing a sampling bias that narrows the variance of the data at each generation, and, in addition to empirical demonstration on GANs and denoising diffusion probabilistic models, provide theoretical analysis for the Gaussian model. Finally, let us mention the study of Bertrand et al. (2023) which sheds light on the critical role of data composition in the stability and effectiveness in generative models, applicable to VAEs (Kingma & Welling, 2014), diffusion models and normalizing flows. They explore scenarios involving a mix of clean data, representative of the true distribution, and synthesized data from previous iterations of the generator. Their analysis reveals that if the data mix consists exclusively of synthesized data, the gen-